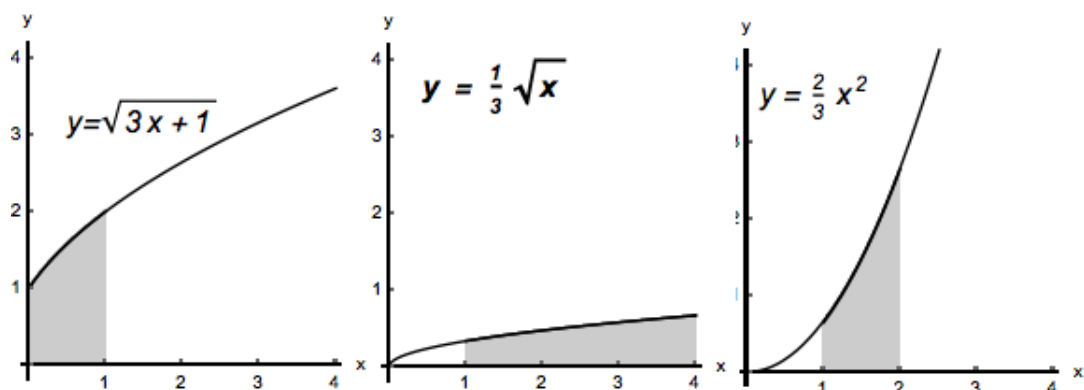


Problem 4

Problem 4

Problem 1 Show that the areas of the three shaded regions in the figure below are equal. Find this area.



Explanation. The area of the first shaded region is given by $\int_0^1 \sqrt{3x+1} dx$. We can evaluate this integral by substituting: $u = 3x+1$, $du = 3 dx$; when $x = 0$, $u = 1$; when $x = 1$, $u = 4$. Therefore $\int_0^1 \sqrt{3x+1} dx = \int_1^4 \frac{1}{3} \sqrt{u} du$ and the last integral gives the area of the region in the middle.

Problem 4

The same integral can also be evaluated by substituting $u = \sqrt{3x+1}$, $du = \frac{3}{2\sqrt{3x+1}} dx = \frac{3}{2u} dx$;
when $x = 0$, $u = 1$; when $x = 1$, $u = 2$.

In this case we have

$$\int_0^1 \sqrt{3x+1} dx = \int_1^2 u \cdot \frac{2}{3} u du = \int_1^2 \frac{2}{3} u^2 du.$$

The last integral gives the area of the third shaded region.
Therefore, the areas are equal.

In order to compute this area, let's evaluate the last integral.

$$\int_1^2 \frac{2}{3} x^2 dx = \frac{2}{3} \left[\frac{x^3}{3} \right]_1^2 = \frac{2}{3} \left[\frac{8-1}{3} \right] = \frac{14}{9}$$